LOCALIZATION OF CABLE FAULTS

TYPES OF CABLE FAULTS

The faults which are mostly likely to occur in the cables are:

1. Ground or Earth Faults

When the insulation of the cable gets damaged, the current starts flowing from core to earth or to cable sheath. Such faults are known as ground or earth faults.

2. Cross or Short-circuit faults

When the insulation between two cables or between two cores of a multi-core cable gets damaged, the current starts flowing from one cable to another cable or from one core to another core of a multicore cable directly (without passing through the load). Such faults are known as short-circuit faults.

3. Open-circuit faults

When the conductor of a cable is broken or joint is pulled out there is no current in the cable. Such faults are known as open-circuit faults.

First the nature of fault is determined and then the point of fault is located. For determination of nature of faults, the insulation resistance of each of each core to ground and between cores is measured with the aid of a megger. The low value of insulation resistance between any core and earth indicates the ground fault whereas the low value of insulation resistance between two cores (with far ends of the cable isolated from load) indicates short-circuit fault.

For determination of open-circuit fault, the far end of the cable, in case of single cable, is earthed (or the far ends of the cables are interconnected), an ammeter is inserted in series with the cable and low voltage supply is connected across the cable and earth or cables. Zero deflection in ammeter indicates open-circuit fault.

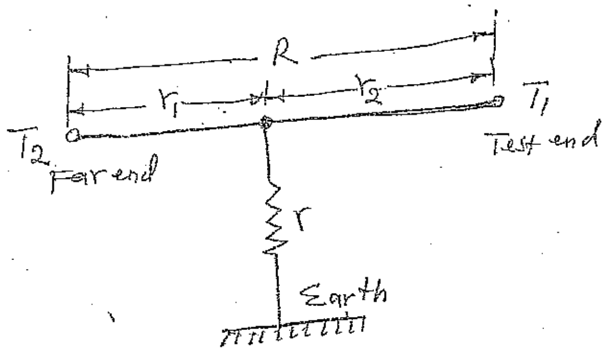
LOCATION OF CABLE FAULTS

There are various methods for locating the point of faults.

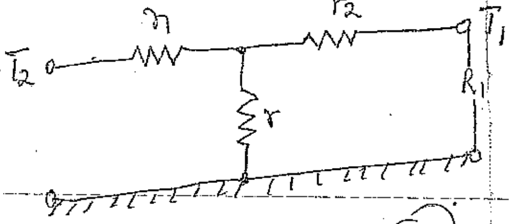
1. BLAVIER'S TEST

Blavier's test is used to locate the ground fault of a single cable i.e. when no other cables run along with the faulty one.

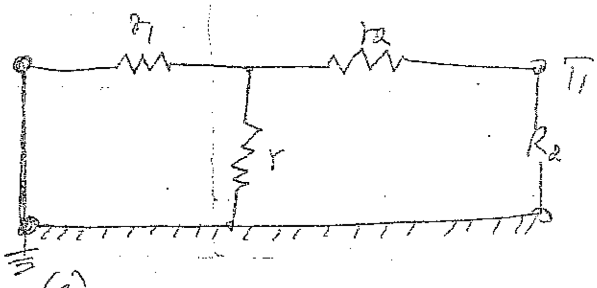
This test is performed with the aid of a low low-voltage supply and either an ammeter or voltmeter meter or a bridge network.



(a)



(b)



(c)

Figure 1: Blavier's test for Ground Fault of a single core cable

In this test resistance between one end of the cable T_1 and earth is measured first with the far end T_2 isolated from earth, as in Figure 1(b), and then with the far end T_2 earthed as shown in Figure 1 (c). Let the two readings be R_1 and R_2 respectively.

If r_1 and r_2 are the conductor resistances of the lengths of cable 'far end' to fault and 'test end' to fault respectively and r is the resistance of fault to earth then

$$R_1 = r_2 + r \qquad \ldots (1)$$

$$\text{and } R_2 = \frac{r_1 r}{r_1 + r} + r_2 \qquad \ldots (2)$$

The total resistance of the conductor is

$$R = r_1 + r_2 \qquad \ldots (3)$$

Substituting $r = R_1 - r_2$ from eqn. (1) in equation (2) we have

$$R_2 = r_2 + \frac{(R_1 - r_2)\, r_1}{(R_1 - r_2) + r_1}$$

$$r_2 = R_2 - \sqrt{(R_1 - R_2)(R_1 - R_2)} \qquad \ldots (4)$$

If the total length of the cable is L meters, the length of the cable between far end and fault is L_1 meters, length of cable between test end and fault is L_2 meters and cross-section of conductor is uniform then

$$\frac{L_1}{L_2} = \frac{r_1}{r_2}$$ Add 1 to both sides

and $$\frac{L_1 + L_2}{L_2} = \frac{r_1 + r_2}{r_2}$$; but $L_1 + L_2 = L$

or then $$\frac{L}{L_2} = \frac{r_1 + r_2}{r_2}$$

$$L_2 = L \times \frac{r_2}{r_1 + r_2} \quad \ldots (5)$$

Thus the distance of fault from test end can be determined.

~~Murray~~ LOOP TESTS

These tests are performed for the location of either an earth fault or a short-circuit fault in underground cables provided that a sound cable runs along with the grounded or a short-circuited cable. In these ~~tests~~ the resistance of fault does not affect the results obtained except when the resistance of fault is very high. There are two loop tests usually used and are known as Murray loop and Varley loop tests. These tests employed the principle of Wheatstone bridge.

1. MURRAY LOOP TEST

The connection diagrams to locate earth fault and short-circuit fault by Murray loop test method are shown in Figures 2 and 3 respectively. Wheatstone bridge principle is used in these tests.

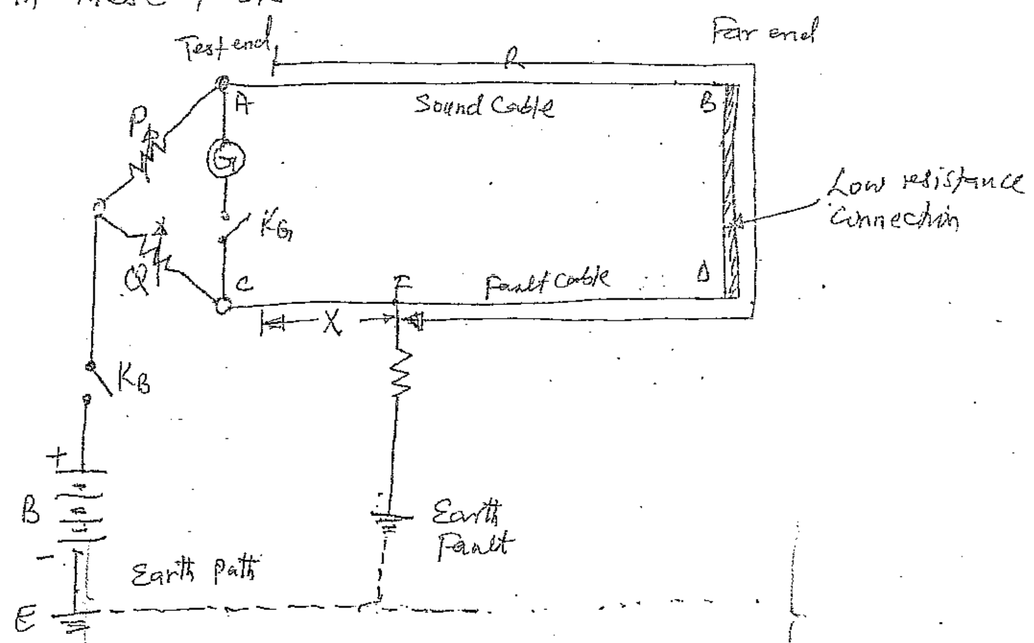


Figure 2: Murray Loop test (Earth fault)

AB is the Sound Cable and CD is the faulty cable; the earth fault occurring at point F

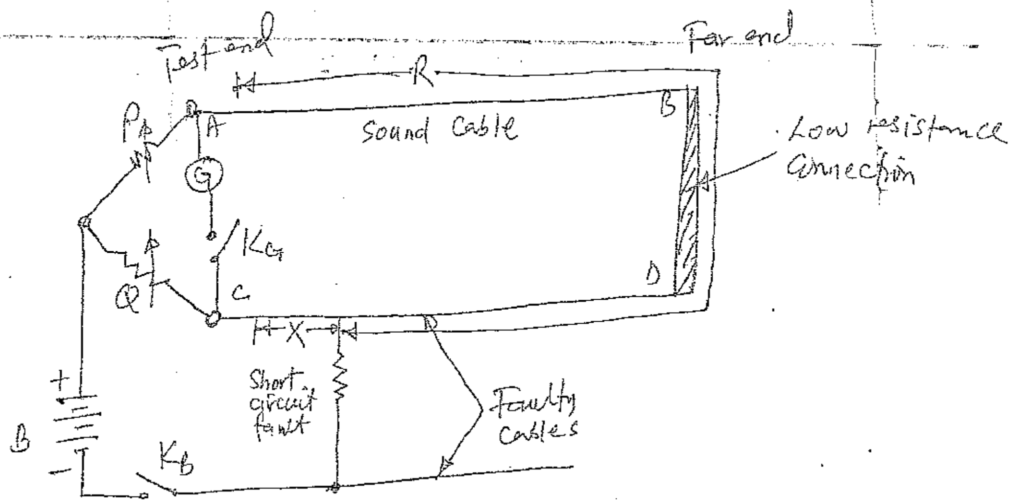


Figure 3: Murray loop test (short circuit fault).

P and Q are the two ratio arms consisting of step (variable) resistors and slide wire, G is the galvanometer, B is the battery, K_B is a battery key and K_G is a galvanometer key. The fault cable is connected to a second cable through low resistance link at the far end. Bridge is balanced by adjusting the resistance of ratio arms P and Q until the galvanometer indicates zero. Let R = resistance of the conductor loop upto the fault from the test end, while X = resistance of the other length of the loop.

In balanced condition of bridge

$$\frac{P}{Q} = \frac{R}{X}$$ Add 1 to both sides

$$\text{or } \frac{P}{Q} + 1 = \frac{R}{X} + 1$$

If r is the resistance of each cable, then $R + X = 2r$

$$\text{or } \frac{P+Q}{Q} = \frac{R+X}{X} = \frac{2r}{X}$$ where $2r$ = twice resistance of one of the cable.

$$\text{or } X = \frac{Q}{P+Q} \times 2r$$

where r is resistance of one of the cables when free from fault. If l is the length of each cable in meters, then

∴ resistance per meter length of cable $= r/l$

∴ Distance of fault from testing end $= \frac{\text{Resistance of the length of loop}}{\text{Resistance per meter length of cable}}$

$$d = \frac{X}{r/l} = \frac{lX}{r} = \frac{Q}{P+Q} \times 2r\frac{l}{r}$$

$$= \frac{Q}{P+Q} \times 2l \text{ meters} \quad \cdots (6)$$

or $= \frac{Q}{P+Q} \times$ (loop length) meters.

The connection for short-circuit test in Figure 3 are practically the same as in earth fault test except, that a portion of one of the short-circuiting cables is used instead of earth as return path in the battery circuit. Balance is obtained as before.

At balance $\frac{P}{Q} = \frac{R}{X}$; Add 1 to both sides gives $\frac{P+Q}{Q} = \frac{R+X}{X}$

$$\text{or} \quad \frac{Q}{P+Q} = \frac{X}{R+X} \quad \text{(inverted)}$$

$$\text{or} \quad X = \frac{Q}{P+Q}(R+X) \quad \cdots (7)$$

Where $(R+X)$ is the total resistance of the loop and is assumed to be known.

Since P, Q and the total resistance of the loop is known so X and hence the distance of fault from lower end of ratio arm Q can be determined.

VARLEY LOOP TEST

In this test also sound cable is required in addition to the existing cable. The circuit diagrams to determine the location of ground fault and short-circuit fault by method of Varley loop tests are shown in Figure 4 and Figure 5 respectively.

The difference between Murray loop and Varley loop tests is that in Varley loop test provision is made for the measurement of total loop resistance instead of obtaining it from the relation $R = \rho \frac{l}{a}$.

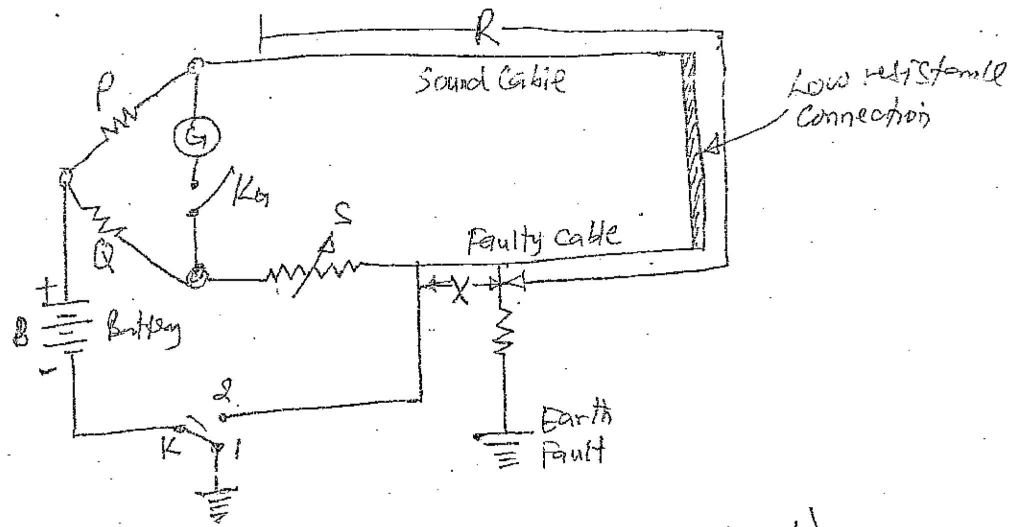


Figure 4: Varley loop test (Earth Fault)

In this test the ratio arms P and Q are fixed and balance is obtained by varying the known variable resistance S. Balance is obtained once the with ~~the~~ key in the battery circuit on stud 1 and then on stud 2.

At balance for earth fault or short-circuit fault with key K on stud 1.

$$\frac{P}{Q} = \frac{R}{X+S_1}$$

Where S_1 is the setting of variable resistance S.

or $$\frac{P+Q}{Q} = \frac{R+X+S_1}{X+S_1}$$ Add 1 to both sides

or $$X = \frac{Q(R+X)-PS_1}{P+Q} \quad \cdots (8)$$

At balance for earth fault or short-circuit fault when key K on stud 2,

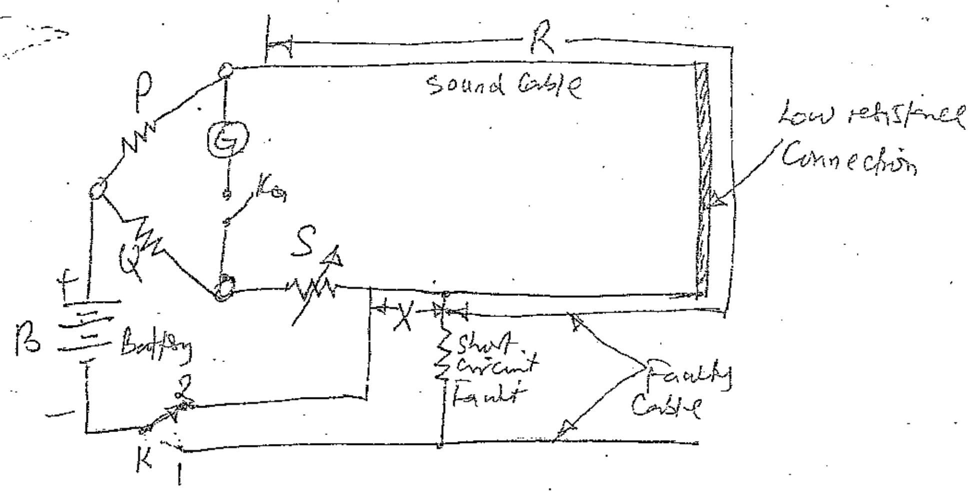


Figure 5: Varley Loop test (Short-circuit)

$$\frac{P}{Q} = \frac{R+X}{S_2}$$

Where S_2 is the new setting of variable resistance S.

or $(R+X)Q = PS_2 \quad \cdots (9)$

or loop resistance $(R+X) = \frac{PS_2}{Q}$

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From equations (8) and (9) we have

$$X = \frac{P(S_2 - S_1)}{P + Q} \quad \ldots (10)$$

Thus X can be determined from the known values of P, Q, S_2 and S_1.

Example 1.

In locating a ground fault by Varley loop test the following observations were recorded:

(i) With key K at stud 1, a balance was obtained when $P = 10$, $Q = 80$ and $S = 2,400\ \Omega$.

(ii) With key K at stud 2, a balance was obtained when $P = 10$, $Q = 80$ and $S = 3,400\ \Omega$.

Determine the distance of the fault from the testing end. Each conductor is 5 km long.

Soln: –

$P = 10\ \Omega$; $Q = 80\ \Omega$; $S_2 = 3,400\ \Omega$; $S_1 = 2400\ \Omega$

$$\therefore X = \frac{P(S_2 - S_1)}{P + Q} = \frac{10(3,400 - 2,400)}{10 + 80} = 111.11\ \Omega$$

Loop resistance, $R + X = \frac{P}{Q} S_2 = \frac{10}{80} \times 3,400$

or $r = 425\ \Omega$

Resistance per km length of cable $= \frac{r}{l} = \frac{425}{10} = 42.5\ \Omega$

$\because l = 2 \times 5 = 10$ i.e. twice loop length

Distance of fault from testing end $d = \frac{X}{r/l} = \frac{111.11}{42.5} = 2.623$ km

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Example 2

In a test by Murray loop for ground fault on 1,000 metres of cable having a resistance of 1·6 Ω per km, the faulty cable is looped with a sound cable of the same length and crossection. If the ratio of the other two arms of testing network at balance are in the ratio of 3 : 1, calculate the distance of the fault from the testing end of the cable.

Soln:–

The ratio of two arms, $\frac{P}{Q} = 3$

or $\frac{P+Q}{Q} = 3 + 1 = 4$

[from add 1 on both side
$\frac{P}{Q} + 1 = \frac{3}{1} + 1$
$\frac{P+Q}{Q} = 3+1$]

Total resistance of loop (R+X) is

$\therefore \quad R + X = \frac{1.6}{1000} \times 2 \times 1000$

$= 3.2\ \Omega$

From eqn (7) $X = \frac{Q}{P+Q}(R+X) = \frac{1}{4} \times 3.2 = 0.8\ \Omega$

Distance of fault from testing end is given by

$\frac{X}{r/l} = \frac{0.8}{1.6/1000} = 500$ metres

Transducer is an electronic device that converts signals from one form to another.

TRANSDUCERS

nstrumentation system: An instrumentation system consists of a
mber of components used to perform ~~a~~ measurement and record the
sult. It consists of three major elements: an input devices, a
gnal conditioning or processing device, and an output device.
e input device receives the quantity under measurement and
elivers a proportional electrical signal to the signal-conditioning
evice. Here the signal is amplified, filtered, or modified to a
ormal acceptable to the output device. The output device may be
Simple indicating meter, an oscilloscope, or a chart recorder for
isual display. It may be ~~a~~ magnetic tape recorder for temporary
permanent storage of the input data, or it may be a digital
ompute for data manipulation or process control.

ransducer- Is a device that converts non-electrical quantities into
n electrical signal. Such non-electrical quantities are heat, light
tensity, humidity. Others such as mechanical force or displacement,

Transducers may be classified according to their application, method
energy conversion, nature of the output signal etc.
Transducers are classified as passive transducers and self generating
ransducers.

ELECTRICAL TRANSDUCERS

lectrical transducers converts non-electrical quantities into electrical
uantities first and then measured.

TYPES OF ELECTRICAL TRANSDUCERS: Various types of Electrical
ransducers as follows:

Resistive type (ii) Capacitive type (iii) Inductive type.

1. Resistive type Transducer - In such a transducer, resistance between the output terminals of a transducer gets varied according to the measurand.

Resistance of any metal conductor is given by the equation

$$R = \frac{\rho l}{a} \quad \ldots (1)$$

Where ρ is the resistivity of the material of conductor in Ω/m,

l is the length of conductor in meters and

a is the cross-sectional area of conductor in m^2.

Physical phenomenon i.e input signal to the transducer causes variation in resistance by changing any one of the quantities ρ, l and a. For measurement of displacement length of conductor is varied in potentiometer thereby resulting in change in resistance. For measurement of force and pressure resistance of a conductor or semiconductor is varied in strain gauge. Variation in temperature causes change in resistivity of the conductor. With some devices resistance varies with the change in light intensity because of photo conductive effect, While with others resistance varies on exposure to magnetic field due to magnetoresistance effect.

2. Inductive transducers - These transducers operate generally upon one of the following three principles

(i) Variation of self inductance of the coil

(ii) Variation of Mutual Inductance of the coil.

(iii) Provision of eddy current.

(i) Transducer operating on the principle of Variation of Self-inductance

The self inductance of a coil is given by the equation

$$L = \frac{N^2}{l/\mu A} = N^2 \frac{\mu A}{l} = N^2 \mu G \quad \ldots (2)$$

where N is the number of turns on the coil,
l is the mean length of the magnetic path.
A is the area of the cross-section of the magnetic path.
μ is the permeability of the magnetic material.
A/l is called the geometric form factor, G.

From the above equation it shows that the self inductance of a coil can be varied by varying the number of turns on the coil, the permeability of the magnetic material or by changing the geometrical configuration of the magnetic circuit.
These transducers are usually used for measurement of displacement both linear and angular displacement.

Transducers operating on the principle of variation of Mutual inductance.
Such transducers operate on the fact that mutual inductance between two coils depends upon the self inductance of the coils and coefficient of coupling between them, as mutual inductance between two coils is given by the equation

$$M = K\sqrt{L_1 L_2} \quad \ldots (3)$$

where K is the coefficient of coupling and L_1 and L_2 are self inductances of the coils.
In such transducers two or more coils are used and the mutual inductance between the coils is varied as per measurand.

Transducers operating on the principle of production of eddy currents.
These type of transducers operate on the principle that when a conducting plate is kept near a coil carrying

alternating current, eddy currents are induced in the conducting plate, producing its own magnetic field in opposition to the main field created by the coil. Thus eddy current induced in the conducting plate reduce the net flux linking with the coil and so the inductance of the coil is reduced. The nearer is the plate to the coil, the higher are the induced eddy currents and so higher is the reduction in the inductance of the coil. Thus, the inductance of the coil changes with the movement of the plates.

3. Capacitive Transducers

The variable capacitance transducer comprises of a capacitor, the capacitance of which is varied by the non-electrical quantity being measured.

The capacitance of a parallel plate capacitor, C is given by an equation

$$C = \frac{\varepsilon_0 \varepsilon_r A}{d} \text{ Farads} \quad --- (4)$$

Where ε_0 is the permittivity of free space and is equal to 8.854×10^{-12} F/M;

ε_r is the relative permittivity of the dielectric,

A is the area of plates in M^2 and

d is the distance between the two plates in metre.

As the capacitance of a capacitor depends upon the area of its electrodes, the distance between them and the relative permittivity of the dielectric material, hence such transducer can be used for measurement of non-electrical quantities by affecting any one of the above mentioned parameters.